



Folic Acid-Derived Low-dimensional carbons for efficient oxidative dehydrogenation of ethylbenzene

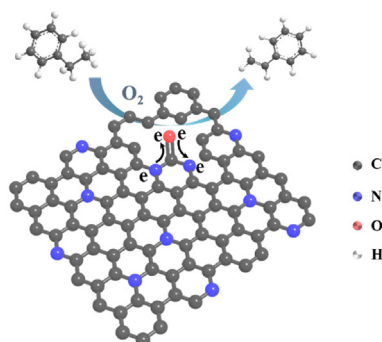
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GRAPHICAL ABSTRACT

Carbon materials with high nitrogen contents were prepared for understanding the influence of nitrogen dopants on the catalytic activity.



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ABSTRACT

The oxidative dehydrogenation (ODH) of alkane is one of the most attractive routes in alkane production because of its favourable thermodynamic characteristic. Nitrogen-doped nanocarbons have demonstrated great potential in this reaction due to its cost-effective, high catalytic activity and stability. However, the influence of nitrogen on the catalytic properties of carbon materials is poorly understood due to the complexities of surface oxygen and nitrogen functional groups. Here we derive the performance descriptor that account for the nitrogen-dependent carbocatalysis in ODH reaction. To achieve this, we designed a set of nitrogen-doped nanocarbon materials with tunable nitrogen species by hydrothermal carbonization (HTC) treatment of the biomass folic acid (FA), which are applied in ODH of ethylbenzene. Among them, FA-180-1000 catalyst can achieve high ethylbenzene conversion (up to ~62 %) and styrene selectivity (~87 %), outperforming other HTC carbon-based catalysts. Structural characterizations and kinetic analyses revealed that nitrogen doping strongly interferes the charge polarization of —C=O site via electron transfer between —C=O , and nitrogen (mainly pyridine nitrogen and graphitic nitrogen) thus enhancing the reactivity of —C=O . Furthermore, the induction period during reaction process can be shortened by applying of sulfuric acid-assisted HTC method for constructing nitrogen-doped carbon catalyst with low crystallinity. The present work provides new insights into the contribution of nitrogen doping to the ODH reaction of carbon nanocatalysts, as well as guidance for the rational design of carbon catalysts for the conversion of hydrocarbons to high-value chemicals.

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